

The Particle Model — Paper A

MEDIUM • NON-CALCULATOR • 35 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided.
- No calculator is needed anywhere on this paper.
- Where a question says **explain**, your answer must talk about **particles** — their arrangement, their movement, the spaces between them, or the forces between them.
- The mark for each question is shown in square brackets on the right.

SECTION 1 – THE THREE STATES

- 1.** Complete the table. Each empty box is worth one mark. [6]

	Solid	Liquid	Gas
Arrangement	regular pattern		
Movement		slide past each other	
Spacing	very close		

- 2.** Name the change of state in each case.

(a) Solid to liquid [1]

(b) Gas to liquid [1]

(c) Liquid to gas [1]

(d) Solid straight to gas [1]

- 3.** Answer these two questions about the properties of the states.

(a) Which state has a fixed volume but no fixed shape? [1]

(b) Which state has neither a fixed volume nor a fixed shape? [1]

SECTION 2 – DIAGRAMS AND EXPLANATIONS

4. In the space below, draw a particle diagram for a solid, a liquid and a gas. Label each one. [3]

5. Explain why a gas can easily be compressed, but a liquid can hardly be compressed at all. [3]

6. State the melting point and the boiling point of water in degrees Celsius. [2]

7. Explain, in terms of particles, why a solid keeps its shape. [2]

- 8.** Changes of state are described as **physical** changes rather than chemical changes. State what this means, and give one piece of evidence for it. [3]

- 9.** 20 g of ice is left in a sealed container until it has all melted. State the mass of water produced, and explain your answer. [2]

- 10.** Give three properties of a gas. [3]

End of paper. Check every answer has a unit or a form the question actually asked for.